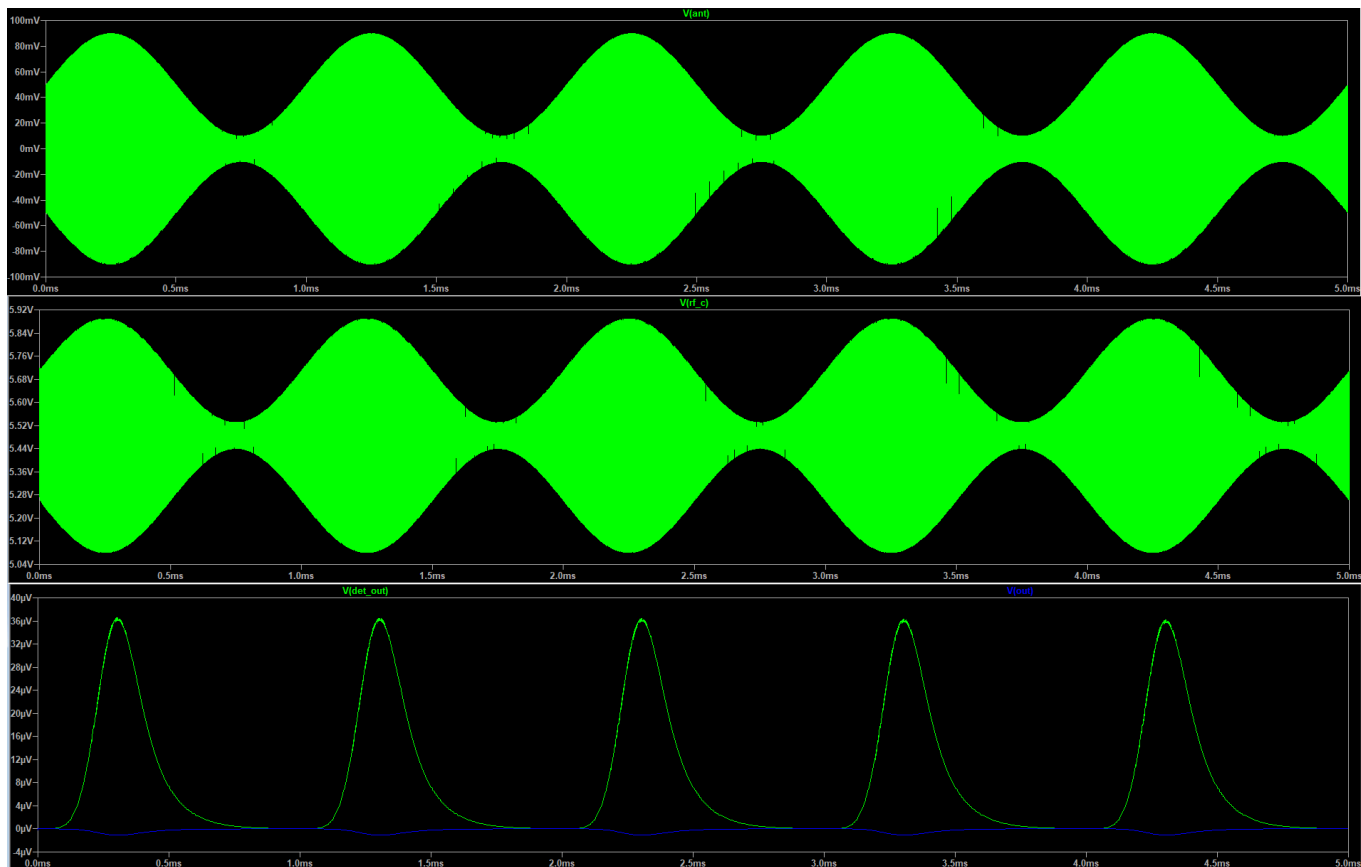


Ideal_LTSpice_Netlist_3

Prompt

I don't understand if what i am observing is a good sign or not, can you help with that ?



Answer

These plots are incredibly helpful for diagnosing the circuit. You are observing a classic rite of passage in radio design: **your signal is hitting a brick wall right in the middle of the circuit.** Here is a breakdown of what your plots are telling you, step-by-step.

The Good News: Your Antenna & RF Stage Work Perfectly

1. **V(ant) (The raw signal):** This is exactly what an AM signal should look like. You have a solid carrier wave, and the top and bottom edges (the envelope) are smoothly undulating with your 1 kHz audio tone. It swings from about -90 mV to +90 mV (roughly 180 mV peak-to-peak).
2. **V(rf_c) (The RF amplifier):** Your first transistor is doing its job! The signal has been amplified. It is now swinging from roughly 5.04V to 5.92V (about 880 mV peak-to-peak). The whole wave is sitting around 5.5V because of the DC bias of the transistor.

The Bad News: The Detector is Dead

3. $V(\text{det_out})$ and $V(\text{out})$: Look closely at the Y-axis on the left side of your third plot. It is measured in μV (microvolts). $V(\text{det_out})$ is peaking at a microscopic $36 \mu\text{V}$ in sharp, distorted spikes, and your final audio $V(\text{out})$ (the blue line) is entirely flat at 0V . Your audio signal has vanished.

The Diagnosis: The Diode "Turn-On" Threshold

To understand why the signal died, we have to look at the component between the RF stage and the detector: the diode.

Even though we used an "ideal" model in the SPICE code, it still mimics a standard Silicon P-N junction. A standard silicon diode (like a 1N4148) requires a forward voltage of about **600 mV to 700 mV** just to "turn on" and let current flow through. Think of it like a heavy door that requires a certain amount of force to push open.

Here is what is happening to your signal:

- The signal leaves the RF amplifier at 880 mV peak-to-peak, riding on a 5.5V DC offset.
- It passes through the coupling capacitor ($C_{\text{RF_OUT}}$). Capacitors block DC. This strips away the 5.5V offset.
- What hits the diode is pure AC, swinging from **-440 mV to $+440 \text{ mV}$** .
- Because $+440 \text{ mV}$ is less than the $\sim 600 \text{ mV}$ threshold required to open the diode, the "door" stays shut. The diode blocks the signal almost entirely, letting only a tiny $36 \mu\text{V}$ trickle leak through.

Because the detector outputs practically nothing, your final audio amplifier has nothing to amplify. To fix this, you either need to boost the gain of your RF amplifier to push the AC swing past 600 mV , or switch to a Germanium diode model (like the 1N34A you mentioned earlier) which has a much lower turn-on voltage of around 200 mV to 300 mV .